

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	23-08-2026

Code of Conduct

I Aswanth N N (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Aswanth N N

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action**, **agent**, **object**, and **destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

Answer:

1. Semantic frame analysis

Query	Action/Frame	Participants/Roles
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER	Customer = Agent, savings account = Source, ₹20,000 = Amount, son's account = Destination
"Block my debit card immediately."	BLOCK	Customer = Agent, debit card = Object
"Send my transaction statement to my email."	SEND	Customer = Agent, transaction statement = Object, email = Destination
"Increase my daily withdrawal limit."	MODIFY	Customer = Agent, withdrawal limit = Object/Target

The semantic frame identifies the **action** being requested and connects it with the entities participating in that action.

For example:

[TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)]

represents not just the words in the sentence but their functional relationships.

2. Incorrect interpretation

The incorrect interpretation is **B4**.

Actual intent:

Increase withdrawal limit

System interpretation:

Decrease withdrawal limit

The system correctly identified the general **MODIFY** operation but incorrectly identified the **direction of modification**.

Therefore, the semantic understanding is incomplete or incorrect.

3. Effect of incorrect semantic roles

Incorrect semantic-role identification can cause an incorrect banking operation.

For example:

“Increase my daily withdrawal limit.”

If the system interprets **increase** as **decrease**, it could submit the wrong transaction to the banking system.

Similarly, confusing:

- source account and destination account
- sender and recipient
- amount and account number

could result in serious financial errors.

Therefore, semantic-role extraction is particularly important in banking because NLP output may directly control financial operations.

4. Recommended frame-based approach

A reliable system should use the following pipeline:

User Query → Intent Detection → Frame Identification → Semantic Role Labeling → Entity Validation → Banking Action

For example:

Transfer ₹20,000 from savings to son's account.

The system should extract:

Frame = TRANSFER

Customer = user

Source = savings account

Amount = ₹20,000

Destination = son's account

It should then validate every required field before executing the operation.

For high-risk operations, the system should also ask for confirmation:

“Transfer ₹20,000 from your savings account to your son's account. Confirm?”

This reduces the risk of incorrect semantic interpretation causing an incorrect transaction.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$
- $\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$
- $\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$
- $\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$
- $\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

Answer:

First-Order Predicate Calculus for Smart Agriculture

1. Representing field observations

The observations can be represented using predicates.

F1

F1 is dry, irrigation is available, and the crop is rice:

$\neg \text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1}) \wedge \text{Rice}(\text{F1})$

F2

$\neg \text{Wet}(\text{F2}) \wedge \text{IrrigationAvailable}(\text{F2}) \wedge \text{Wheat}(\text{F2})$

F3

$\neg \text{Dry}(\text{F3}) \wedge \neg \text{IrrigationUnavailable}(\text{F3}) \wedge \text{Maize}(\text{F3})$

F4

$\neg \text{Wet}(\text{F4}) \wedge \neg \text{IrrigationUnavailable}(\text{F4}) \wedge \text{Rice}(\text{F4})$

2. Determine which fields require irrigation

F1

We have:

$\neg \text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1})$

Using:

$\neg \text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$

we obtain:

$\neg \text{NeedsWater}(\text{F1})$

Then:

$\neg \text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$

Therefore:

$\boxed{\text{Irrigate}(\text{F1})}$

Because F1 is rice:

$\neg \text{Rice}(\text{F1}) \wedge \text{NeedsWater}(\text{F1})$

also gives:

$\neg \text{PriorityIrrigation}(\text{F1})$

So F1 requires **priority irrigation**.

F2

F2 is wet:

$\neg \text{Wet}(\text{F2})$

Using:

$\neg \text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$

we obtain:

$\neg \neg \text{NeedsWater}(\text{F2})$

Therefore:

$\boxed{\text{F2 does not require irrigation}}$

F3

F3 is dry:

$\neg \text{Dry}(\text{F3})$

But irrigation is unavailable:

$\neg \text{IrrigationUnavailable}(\text{F3})$

The first rule requires both:

$\neg \text{Dry}(x) \wedge \text{IrrigationAvailable}(x)$

Since irrigation is unavailable, we cannot derive:

$\neg \text{NeedsWater}(F3)$

Also:

$\text{IrrigationUnavailable}(F3) \rightarrow \neg \text{Irrigate}(F3)$

Therefore:

$\neg \text{Irrigate}(F3)$

F4

F4 is wet:

$\text{Wet}(F4)$

Therefore:

$\neg \text{NeedsWater}(F4)$

Also irrigation is unavailable:

$\neg \text{Irrigate}(F4)$

So:

$\text{F4 does not require irrigation}$

Final result

Field	Needs Water	Irrigation	Priority
F1	Yes	Yes	Yes
F2	No	No	No
F3	Unknown/Not derivable	No	No
F4	No	No	No

3. Should F1 and F3 be treated differently?

Yes.

F1:

- Soil = Dry
- Irrigation = Available
- Crop = Rice
- Needs water can be logically derived.
- Irrigation can be performed.
- It receives priority.

F3:

- Soil = Dry
- Irrigation = Unavailable
- Crop = Maize
- Irrigation cannot be performed.

Thus:

$F1 \rightarrow \text{Irrigate}$

while:

$F3 \rightarrow \neg \text{Irrigate}$

The important distinction is that **F3's dry condition does not mean the controller can irrigate it**, because irrigation is unavailable.

4. Is predicate logic alone sufficient?

No.

Classical predicate logic works well when information is:

- complete
- precise

- certain
- correctly measured

Real agricultural sensors can be uncertain.

For example, a soil-moisture sensor might report:

Soil moisture = 29%

The system may not know with certainty whether the field should be classified as **Dry** or **Wet**.

Sensor failures can also produce missing information.

Predicate logic alone generally represents facts as true or false. It does not naturally represent:

- probability
- sensor confidence
- noisy measurements
- conflicting sensor readings
- missing information

Therefore, practical agricultural systems should combine logical rules with methods such as **fuzzy logic, probabilistic reasoning, Bayesian models, or machine learning**.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

Answer:

Word Sense Disambiguation in a Travel Recommendation System

1. Intended senses

Query	Intended Sense	Evidence
Amazon tour	Amazon rainforest	User viewed rainforest trekking packages
Safari booking	Wildlife tour	User selected wildlife resort packages
Java vacation	Java island	User viewed hotels in Bali and Java
Cruise package	Ship journey	User viewed Mediterranean ship packages
Safari download for Windows	Web browser	User clicked a browser download page

The last example is particularly important.

The word **Safari** normally has multiple meanings, but:

“download for Windows”

is a strong cue for the **Apple Safari web browser**, rather than a wildlife tour.

2. Lexical and contextual cues

Amazon

Possible meanings:

- Amazon rainforest
- Amazon company

Cue:

tour, rainforest, trekking

→ rainforest.

Safari

Possible meanings:

- wildlife tour
- web browser

Cue:

booking, wildlife resort

→ wildlife tour.

But:

download, Windows

→ web browser.

Java

Possible meanings:

- Java island
- Java programming language

Cue:

vacation, hotels, Bali

→ Java island.

Cruise

Possible meanings:

- ship journey
- Cruise software/project

Cue:

Mediterranean, packages, ship

→ cruise journey.

Thus WSD uses both **local words** and **broader user context**.

3. Why the same word can have different meanings

Meaning depends on context.

For example:

Safari booking

suggests a wildlife experience.

But:

Safari download for Windows

strongly suggests the browser.

The system therefore cannot assign one fixed meaning to every occurrence of a word.

The same word can have different meanings depending on:

- surrounding words
- search intent
- previous searches
- clicked results
- user preferences
- geographical context
- browsing behaviour

This is why WSD needs contextual information rather than simple dictionary matching.

4. Industrial-scale WSD strategy

A large travel recommendation system can use a multi-stage architecture:

Query → Candidate Sense Generation → Context Encoding → User History → Behaviour Analysis → Sense Ranking → Recommendation

Step 1: Query context

Extract important terms surrounding the ambiguous word.

For example:

Safari + download + Windows

strongly indicates browser.

Step 2: Embeddings

Use contextual embeddings from transformer-based language models.

The representation of **Safari** changes depending on its surrounding words.

Therefore:

Safari + resort

and:

Safari + Windows download

produce different contextual representations.

Step 3: User history

Previous interactions can provide additional evidence.

If a user frequently searches:

wildlife resorts

Kenya safari

African tours

then **Safari** is more likely to mean a wildlife tour.

Step 4: Click behaviour

Clicks provide implicit feedback.

For example:

Safari booking → wildlife resort clicked

strengthens the wildlife-tour interpretation.

Step 5: Sense ranking

The system can combine evidence:

$$[\text{Score}(s) = w_1 \text{Context} + w_2 \text{Embedding} + w_3 \text{History} + w_4 \text{Clicks}]$$

The sense with the highest score becomes the selected interpretation.

This approach is more reliable than using only a dictionary or only the current query.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Answer:

Syntax-Driven Semantic Analysis in Legal Documents

1. Active and passive constructions

Syntax determines the grammatical position of entities, but grammatical position does not always equal semantic role.

Active sentence

The supplier delivered the equipment to the buyer.

Structure:

Supplier → delivered → equipment → to buyer

Semantic roles:

- Supplier = Agent
- Equipment = Object/Theme
- Buyer = Recipient

This interpretation is correct.

Second sentence

The buyer received the equipment from the supplier.

The given interpretation says:

Buyer = Agent, Equipment = Object, Supplier = Recipient

The first two are reasonable in a broad semantic-role analysis:

- Buyer = Recipient
- Equipment = Object/Theme
- Supplier = Source

But calling Buyer an Agent and Supplier a Recipient is incorrect.

The better representation is:

- Buyer = Recipient
- Equipment = Object/Theme
- Supplier = Source

Therefore, this interpretation is incorrect.

Third sentence

The company terminated the contract after the violation.

Roles:

- Company = Agent
- Contract = Object/Theme
- Violation = temporal/circumstantial event

The given interpretation is correct.

Fourth sentence

The contract was terminated by the company.

This is passive voice.

The system incorrectly says:

Contract = Agent
Company = Object

The correct roles are:

- Contract = Object/Theme
- Company = Agent

The word by introduces the agent in a passive construction.

2. Incorrect assignments

There are two problematic interpretations.

Sentence 2

Given:

Buyer = Agent, Supplier = Recipient

Correct:

\[Buyer=Recipient \]\[Supplier=Source \]

Sentence 4

Given:

Contract = Agent, Company = Object

Correct:

\[Contract=Object/Theme \]\[Company=Agent \]

Corrected table

Sentence	Correct Semantic Roles
Supplier delivered equipment to buyer	Supplier = Agent; Equipment = Object; Buyer = Recipient
Buyer received equipment from supplier	Buyer = Recipient; Equipment = Object; Supplier = Source
Company terminated contract	Company = Agent; Contract = Object
Contract was terminated by company	Contract = Object; Company = Agent

3. Why confusing subject and agent is dangerous

A grammatical subject is not necessarily the semantic agent.

This becomes obvious in passive sentences.

The contract was terminated by the company.

Grammatical subject:

contract

Semantic agent:

company

If a legal NLP system simply assumes:

\[Subject = Agent \]

it would incorrectly conclude:

Contract terminated something.

This could seriously affect legal information extraction.

For example, an automated system might incorrectly extract:

Agent = Contract

Action = terminate

Object = Company

instead of:

Agent = Company

Action = terminate

Object = Contract

Such errors could affect:

- contract analysis
- liability extraction
- obligation detection
- dispute analysis
- legal search
- compliance systems

4. Recommended syntax-driven architecture

A robust legal NLP system should use:

Legal Document



Sentence Segmentation



Tokenization



POS Tagging



Dependency/Constituency Parsing



Syntactic Role Detection



Semantic Role Labeling



Entity Linking



Coreference Resolution



Legal Knowledge Graph / Structured Output

Active voice handling

For:

The company terminated the contract.

Dependency analysis identifies:

company → subject

terminated → verb

contract → object

Semantic interpretation:

Company = Agent

Contract = Object

Passive voice handling

For:

The contract was terminated by the company.

The parser identifies:

contract → passive subject

company → by-phrase

Semantic-role rules then map:

contract → Object

company → Agent

Prepositional phrases

The system should also distinguish:

- to → possible Recipient
- from → possible Source
- by → possible Agent in passive constructions
- with → possible Instrument
- for → possible Beneficiary/Purpose

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	4	4	3	4	19	95
2	4	4	4	3	4	19	
3	4	4	4	3	4	19	
4	4	4	4	3	4	19	

Faculty In-charge	Course Coordinator	Program Director
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Date: _____	Date: _____	Date: _____